SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code: ITA03	Course Title: Mobile computing
CO Assessed: CO1	Assessment: Assessment Tool 2 – Short Answers
Weightage: 30 % of CIA	Total Marks: 100 Marks
Student Name:	Reg. No.:
Section / Batch:	Date:

CO1 Statement: Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.

Code of Conduct

I L.Vishnu Vardhan Reddy (192372138). (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: L.Vishnu Vardhan Reddy

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Given: Total spectrum = 45 MHz; each simplex channel = 25 kHz.
Concept / Formula: Number of simplex channels = Total bandwidth / Channel bandwidth. A full-duplex voice channel needs two simplex channels (one forward, one reverse).
Solution: 45 MHz = 45,000 kHz. Simplex channels = 45,000 / 25 = 1800. Full-duplex channels = 1800 / 2 = 900.
Answer: 900 full-duplex voice channels.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Given: Total channels = 560; cluster size N = 7.
Concept / Formula: Channels per cell = Total channels / Cluster size (each cell in a cluster gets an equal share, and the cluster's channels are reused in other clusters).
Solution: 560 / 7 = 80.
Answer: 80 channels per cell.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Given: Carrier spacing = 200 kHz; number of carriers = 100; time slots per carrier = 8.
Concept / Formula: In a TDMA/FDMA system like GSM, total channels = Number of carriers × Time slots per carrier.
Solution: 100 × 8 = 800 (the 200 kHz spacing only fixes the spectrum used, i.e. 100 × 200 kHz = 20 MHz).
Answer: 800 simultaneous voice channels.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

Given: Base station power = 80 W; handset power = 4 W.
Concept / Formula: Power ratio in dB = 10·log10(P1 / P2).
Solution: Ratio = 80 / 4 = 20. 10·log10(20) = 10 × 1.301 = 13.01 dB.
Answer: Power control ratio ≈ 13 dB.

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Given: Carrier (serving) signal C = -60 dBm; interference I = -92 dBm.
Concept / Formula: When both quantities are in dBm, C/I (dB) = C (dBm) − I (dBm).
Solution: C/I = -60 − (-92) = 32 dB. This is a healthy value, since GSM typically needs about 9–12 dB.
Answer: C/I = 32 dB (good link quality).

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Given: Spectrum = 42 MHz; channel bandwidth = 300 kHz; cluster size N = 7.
Concept / Formula: Total channels = Spectrum / Channel bandwidth. Channels per cell = Total channels / N.
Solution: 42 MHz = 42,000 kHz. Total channels = 42,000 / 300 = 140. Channels per cell = 140 / 7 = 20.
Answer: 20 channels per cell.

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Given: Pt/Pr = 64 W at reference distance d; new distance = 2d; path-loss exponent n = 4.
Concept / Formula: Received power varies as Pr ∝ 1 / d^n, so Pr2 = Pr1 × (d1/d2)^n.
Solution: Pr2 = 64 × (1/2)^4 = 64 × (1/16) = 4 W.
Answer: Received power = 4 W (a 12 dB drop).

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Given: Radius is doubled (R2 = 2R1); the same SNR must be kept at the cell edge; inverse-square law (n = 2).
Concept / Formula: Received power ∝ Pt / R^n. To keep received power constant, Pt must scale with R^n.
Solution: Pt2 / Pt1 = (R2 / R1)^2 = 2^2 = 4.
Answer: Transmit power must be increased by a factor of 4 (about +6 dB).

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Given: Cluster size N = 4; channels per cell = 15.
Concept / Formula: Channels per cluster = Cluster size × Channels per cell.
Solution: 4 × 15 = 60.
Answer: 60 channels in the cluster.

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Given: Speed v = 100 km/h; carrier frequency f = 900 MHz; c = 3 × 10^8 m/s.
Concept / Formula: Maximum Doppler shift fd = v / λ = (v × f) / c, with v in m/s.
Solution: v = 100 × 1000 / 3600 = 27.78 m/s. λ = c/f = 3×10^8 / 900×10^6 = 0.333 m. fd = 27.78 / 0.333 = 83.3 Hz.
Answer: Doppler shift ≈ 83 Hz.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Given: Frame duration = 5 ms; slots per frame = 8.
Concept / Formula: Slot duration = Frame duration / Number of slots.
Solution: 5 ms / 8 = 0.625 ms (625 µs).
Answer: 0.625 ms per slot.

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Given: Allocated bandwidth = 24 MHz; carrier spacing = 200 kHz.
Concept / Formula: Number of carriers = Total bandwidth / Carrier spacing.
Solution: 24 MHz = 24,000 kHz. 24,000 / 200 = 120.
Answer: 120 carrier frequencies.

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Given: Transmit power = 8 W; path loss = 33 dB.
Concept / Formula: Convert watts to dBm: P(dBm) = 10·log10(P in mW). Then Received power (dBm) = Transmit power (dBm) − Path loss (dB).
Solution: 8 W = 8000 mW, so P = 10·log10(8000) = 39.03 dBm. Received = 39.03 − 33 = 6.03 dBm.
Answer: Received power ≈ 6 dBm (about 4 mW).

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Given: Frequency f = 1900 MHz; distance d = 2.5 km; reference distance d0 = 1 m.
Concept / Formula: Free-space path loss: FSPL (dB) = 20·log10(d in km) + 20·log10(f in MHz) + 32.44. This is equivalent to the Friis free-space model referenced to d0 = 1 m.
Solution: 20·log10(2.5) = 7.96 dB. 20·log10(1900) = 65.58 dB. FSPL = 7.96 + 65.58 + 32.44 = 105.98 dB.
Answer: Free-space path loss ≈ 106 dB.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Given: Cells per cluster = 6; calls per cell = 250.
Concept / Formula: Cluster capacity = Number of cells × Calls per cell.
Solution: 6 × 250 = 1500.
Answer: 1500 simultaneous calls.

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Given: Call arrival rate λ = 4000 calls/s; average holding time = 3 minutes = 180 s.
Concept / Formula: Little's law (traffic intensity): Active calls = Arrival rate × Average holding time.
Solution: 4000 × 180 = 720,000.
Answer: 720,000 calls active at one time.

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Given: Number of channels = 150; each channel = 25 kHz.
Concept / Formula: Total bandwidth = Number of channels × Bandwidth per channel.
Solution: 150 × 25 kHz = 3750 kHz = 3.75 MHz.
Answer: 3.75 MHz.

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Given: Guard band = 40 kHz; number of bands = 30.
Concept / Formula: Total guard bandwidth = Number of guard bands × Width of each guard band.
Solution: 30 × 40 kHz = 1200 kHz = 1.2 MHz. (If only the 29 gaps between 30 bands are counted, it would be 1.16 MHz.)
Answer: 1.2 MHz wasted in guard bands.

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Given: Speed = 90 km/h; cell radius = 1.5 km; time = 1 hour.
Concept / Formula: Distance travelled = speed × time. Cell crossing distance = diameter = 2R. Handoffs = Distance / Crossing distance.
Solution: Diameter = 2 × 1.5 = 3 km. Distance = 90 × 1 = 90 km. Handoffs = 90 / 3 = 30.
Answer: About 30 handoffs per hour.

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Given: Regional centers = 8; base stations per center = 40; users per base station = 250.
Concept / Formula: Total capacity = Regional centers × Base stations per center × Users per base station.
Solution: 8 × 40 = 320 base stations. 320 × 250 = 80,000.
Answer: 80,000 users.

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Given: Region area = 400 km^2; cell radius R = 400 m = 0.4 km; cell area ≈ R^2.
Concept / Formula: Number of cells = Total area / Area per cell.
Solution: Cell area = $(0.4)^2$ = 0.16 km^2. Cells = 400 / 0.16 = 2500.
Answer: About 2500 cells.

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Given: Channel bandwidth = 30 kHz; total spectrum = 6 MHz.
Concept / Formula: FDMA channels = Total spectrum / Channel bandwidth.
Solution: 6 MHz = 6000 kHz. 6000 / 30 = 200.
Answer: 200 channels.

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Given: Subscribers = 1200; calls per subscriber per hour = 4; average call duration = 2.5 minutes.
Concept / Formula: Offered traffic A (Erlangs) = (Total calls per hour × Average holding time in minutes) / 60.
Solution: Total calls/hour = 1200 × 4 = 4800. A = 4800 × 2.5 / 60 = 12,000 / 60 = 200.
Answer: 200 Erlangs.

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Given: Total bandwidth = 40 MHz; bandwidth per user = 400 kHz (the 2.5 GHz carrier does not change the count).
Concept / Formula: Maximum users = Total bandwidth / Bandwidth per user.
Solution: 40 MHz = 40,000 kHz. 40,000 / 400 = 100.
Answer: 100 simultaneous users.

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Given: Users per base station = 600; number of base stations = 1200.
Concept / Formula: Network capacity = Users per base station × Number of base stations.
Solution: 600 × 1200 = 720,000.
Answer: 720,000 users.

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Given: Cells paged per round = 4; total cells = 200.
Concept / Formula: Worst case, the subscriber is in the last group, so paging rounds = Total cells / Cells per round.
Solution: 200 / 4 = 50.
Answer: 50 paging rounds (worst case).

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Given: Capacity = 100,000 connections; call attempts = 40,000/min; holding time = 3 min.
Concept / Formula: Offered traffic = Arrival rate × Holding time. Blocking fraction ≈ (Offered − Capacity) / Offered (simple approximation, not Erlang-B).
Solution: Offered = 40,000 × 3 = 120,000 Erlangs. Excess = 120,000 − 100,000 = 20,000. Fraction = 20,000 / 120,000 = 0.1667.
Answer: About 16.7% of the offered traffic is blocked.

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Given: Subscribers = 600,000; roaming share = 15%.
Concept / Formula: Roaming subscribers = Percentage × Total subscribers.
Solution: 0.15 × 600,000 = 90,000.
Answer: 90,000 subscribers roaming.

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Given: Capacity = 12,000 calls/s; current load = 9,000 calls/s.
Concept / Formula: Utilization (%) = (Current load / Capacity) × 100.
Solution: (9000 / 12,000) × 100 = 0.75 × 100 = 75.
Answer: 75% utilization.

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Given: Reuse factor (cluster size) N = 4; spectrum = 48 MHz; channel = 200 kHz.
Concept / Formula: Total channels = Spectrum / Channel width. Channels per cell = Total channels / N.
Solution: 48 MHz = 48,000 kHz. 48,000 / 200 = 240. 240 / 4 = 60.
Answer: 60 channels per cell.

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Given: Allocation = 30 MHz; channel spacing = 200 kHz.
Concept / Formula: Number of channels = Total bandwidth / Channel spacing.
Solution: 30 MHz = 30,000 kHz. 30,000 / 200 = 150.
Answer: 150 channels.

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Given: Handoff attempts = 2500; success rate = 99%.
Concept / Formula: Successful handoffs = Success rate × Attempts.
Solution: 0.99 × 2500 = 2475 (so 25 handoffs failed).
Answer: 2475 successful handoffs.

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Given: Transmit power = 12 W; path loss = 45 dB.
Concept / Formula: Pt (dBm) = 10·log10(P in mW). Received power (dBm) = Pt (dBm) − Path loss (dB).
Solution: 12 W = 12,000 mW, so Pt = 10·log10(12,000) = 40.79 dBm. Pr = 40.79 − 45 = -4.21 dBm.
Answer: Received power ≈ -4.2 dBm (about 0.38 mW).

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Given: Cells per cluster = 7; users per cell = 900.
Concept / Formula: Cluster capacity = Number of cells × Users per cell.
Solution: 7 × 900 = 6300.
Answer: 6300 users.

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Given: Frame duration = 4.8 ms; slots per frame = 8.
Concept / Formula: Slot duration = Frame duration / Number of slots.
Solution: 4.8 ms / 8 = 0.6 ms (600 µs).
Answer: 0.6 ms per slot.

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Given: Speed v = 150 km/h; frequency f = 1.8 GHz; c = 3 × 10^8 m/s.
Concept / Formula: Maximum Doppler shift fd = v / λ = (v × f) / c.
Solution: v = 150 × 1000 / 3600 = 41.67 m/s. λ = 3×10^8 / 1.8×10^9 = 0.1667 m. fd = 41.67 / 0.1667 = 250 Hz.
Answer: Doppler shift = 250 Hz.

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Given: Capacity = 250,000 calls; call rate = 70,000/min; average duration = 4 min.
Concept / Formula: Offered load = Arrival rate × Holding time. Blocked share ≈ (Offered − Capacity) / Offered.
Solution: Offered = 70,000 × 4 = 280,000 Erlangs. Excess = 280,000 − 250,000 = 30,000. Share = 30,000 / 280,000 = 0.107.
Answer: About 10.7% of the offered load is blocked.

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Given: Cluster size = 9 cells; users per cell = 600.
Concept / Formula: Cluster capacity = Cells per cluster × Users per cell.
Solution: 9 × 600 = 5400.
Answer: 5400 users.

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Given: Calls processed = 60,000/s; drop rate = 0.4% = 0.004.
Concept / Formula: Dropped calls per second = Drop rate × Calls processed per second.
Solution: 0.004 × 60,000 = 240.
Answer: 240 calls dropped per second.

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Given: Transmit power = 43 dBm; received power at cell edge = -87 dBm (antenna gains ignored).
Concept / Formula: Path loss (dB) = Transmitted power (dBm) − Received power (dBm).
Solution: PL = 43 − (-87) = 43 + 87 = 130.
Answer: Path loss = 130 dB.

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Given: Area per base station = 2.5 km²; number of stations = 150.
Concept / Formula: Total coverage area = Area per station × Number of stations (assuming no overlap).
Solution: 2.5 × 150 = 375.
Answer: 375 km² total coverage.

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Given: Reuse factor = 6; spectrum = 48 MHz; channel = 200 kHz.
Concept / Formula: Total channels = Spectrum / Channel width. Channels per cell = Total channels / Reuse factor.
Solution: 48,000 kHz / 200 kHz = 240. 240 / 6 = 40.
Answer: 40 channels per cell.

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Given: Users per paging cycle = 6000; total subscribers = 150,000.
Concept / Formula: Paging cycles = Total subscribers / Users per cycle.
Solution: 150,000 / 6000 = 25.
Answer: 25 paging cycles.

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Given: Call duration = 9 minutes; one handoff roughly every 3 minutes.
Concept / Formula: Number of handoffs = Call duration / Time between handoffs.
Solution: 9 / 3 = 3.
Answer: About 3 handoffs during the call.

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Given: Signal = -85 dBm; noise floor = -115 dBm.
Concept / Formula: SNR (dB) = Signal (dBm) − Noise (dBm).
Solution: SNR = -85 − (-115) = 30.
Answer: SNR = 30 dB.

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Given: Users = 250; guaranteed rate per user = 150 kbps.
Concept / Formula: Required aggregate rate = Users × Rate per user.
Solution: 250 × 150 = 37,500 kbps = 37.5 Mbps (assuming 1 bit/s/Hz it would need 37.5 MHz).
Answer: Minimum capacity of 37.5 Mbps.

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Given: Subscribers = 1500; calls per subscriber per hour = 4; call length = 2.5 min.
Concept / Formula: Traffic A (Erlangs) = (Calls per hour × Holding time in minutes) / 60.
Solution: Calls per hour = 1500 × 4 = 6000. A = 6000 × 2.5 / 60 = 15,000 / 60 = 250.
Answer: 250 Erlangs.

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Given: Control channels = 250; width of each = 12 kHz.
Concept / Formula: Total bandwidth = Number of channels × Bandwidth per channel.
Solution: 250 × 12 kHz = 3000 kHz = 3 MHz.
Answer: 3 MHz used by control channels.

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Given: Connections = 1500; average rate = 600 kbps each.
Concept / Formula: Total throughput = Connections × Rate per connection.
Solution: 1500 × 600 = 900,000 kbps = 900 Mbps.
Answer: Total throughput = 900 Mbps.

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

Given: Tower radius R = 4 km; circular coverage area = πR^2; service region = 350 km².
Concept / Formula: Number of towers = Region area / Area per tower (rounded up to the next whole tower).
Solution: Area per tower = $\pi \times 4^2 = 3.1416 \times 16 = 50.27$ km². Towers = 350 / 50.27 = 6.96, which rounds up to 7.
Answer: About 7 towers.